Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[18AU53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

INTERNAL COMBUSTION ENGINES

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List the different important properties of fuels related to combustion. Explain four of them in detail.	**08**
	b.	What is meant by scavenging in IC engines? Impact of Scavenging in 2-stroke and 4-stroke cycle engines.	**06**
2.	**a.**	State the merits and demerits of fuel injection system used in SI engines.	**06**
	b.	Enumerate types of combustion chambers used in SI Engine. Discuss the functioning of any two combustion chambers.	**08**
3.	**a.**	What is meant be knocking in CI engines? List the factors affecting the knocking	**04**
	b.	Write short notes on i) Swirl combustion chamber ii) Squish combustion chamber	**10**
		OR	
4.	**a.**	Name the various types of injectors used in diesel injection system,	**04**
	b.	Explain the stages of combustion in CI engines with P-θ diagram.	**10**
5.	**a.**	Write Short notes on Indian emission norms	**06**
	b.	Explain the formation of CO_x, NO_x and Soot in the diesel engine emissions	**08**
		OR	
6.	**a.**	Explain the working of 3-way catalytic convertor with neat sketch	**14**
7.	**a.**	Write short notes on: i) Stratified charge engine ii) HCCI Engine	**10**
	b.	State the requirements for the alternate fuels in IC engines	**04**
		OR	
8.	**a.**	Explain the merits and demerits of alcohol as fuel in IC engines	**06**
	b.	Short notes on: i) Sources of Biodiesel ii) Process of transesterification	**08**